

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Identifies Donovan	AO1.2	B1	First Outlier Donovan
	Infers a reason for Donovan - a data entry error has been made - Donovan gets nervous/stressed under exam conditions and performed poorly in most recent test - Donovan is not a very good player despite a lot of practice - Just started playing so practised longer hours but performed poorly in exam - External factors/illness Accept other reasonable reason linking Donovan's practice time to their performance in the exam.	AO2.2b	E1	Reason A data entry error has been made (should be 5 not 50)
	Identifies Collins	AO1.2	B1	
	Infers a reason for Collins - a data entry error has been made - naturally very good piano player so does little practice 'Lucky' test score Accept other reasonable reason linking Collins's practice time to their performance in the exam.	AO2.2b	E1	Second outlier Collins Reason Naturally very able student.
(b)				
(i)	Describes correlation correctly, at least strong positive. Accept non-linear correlation, but do not accept numerical value to indicate strength	AO2.5	B1	Strong Positive Correlation
(ii)	Interprets correlation in	AO3.2a	E1	Students who complete

context (as given in typical solution OE) Do not accept the better you do in the exam the more you practised			more practice perform better in the exam
Total 6 marks			

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Explains that Bilal may be wrong with reference to rounding	AO2.4	E1	Bilal's statement is incorrect because the figures given in the dataset are rounded to integers and therefore the actual values may total to a rounded value that is not the total of the two component rounded values.
(b)	States that Maria's claim is not supported Explains that the actual recorded consumption values for Takeaway confectionery are non-zero with reference to knowledge of the Large Data Set... OR Just because the values happen to be 0 in these four periods values will not necessarily always be 0, with reference to knowledge of the Large Data Set.	AO2.4	E1	Maria's claim is not supported because the actual data values, not the zeros that appear in the table, for the consumption of Takeaway confectionery, are not equal to zero. Maria needs to understand that when using the LDS spreadsheet the decimal values are visible (to over ten decimal places), but that the summary data shown here is rounded to the nearest integer.
Total 2 marks				

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Identifies likely outlier	AO1.2	B1	East region 2007

(i)				
(ii)	Finds Q_1 and Q_3 for East region FT NW if that was their outlier in part (a)(i)	AO1.1b	B1F	For East region $Q_1 = 0$ $Q_3 = 4$ (For NW $Q_1 = 2$ $Q_3 = 6$)
	Finds IQR for East region FT NW if that was their outlier in part (a)(i)	AO1.1b	B1	$IQR = 4$ E ($IQR = 4$ NW)
	Completes argument that uses formula given in the question together with 'their' values found for Q_1 , Q_3 and IQR and 'their' outlier for (a)(i) to confirm that it is an outlier Award credit here provided 'their' values do confirm 'their' identified value to be an outlier	AO2.1	R1	$4 + 1.5 \times 4 = 10$ $21 > 10$ Hence East 2007 is an outlier
(b)	Explains reason for mean being unrepresentative	AO2.4	E1	Mean would be unrepresentative as it would be affected by the large value outlier.
(c)	Provides explanation	AO2.2b	E1	Allow: Error in data entry; Some event in 2007 led to an increase in dried milk products consumption locally; Disease hit dairy herds.
Total 6 marks				

Q4.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Indicates that 'Semi-skimmed milk' is part of the whole 'Skimmed milks' category.	AO2.4	E1	In Figure 2, the vertical axis represents 'Skimmed milks' whereas in Figure 1 the vertical axis represents 'Semi-skimmed milk' that is just one part of the 'Skimmed milks' category. Hence, for each region,

				the recorded values for 'Semi-skimmed milk' will always be lower than the ones for 'Skimmed milks'.
(b)	States: strong positive correlation between purchases of ' Semi-skimmed milk ' and ' Liquid wholemilk, full price ' for most regions.	AO2.5	B1	Regions A, B, D, F, G, H and J indicate strong positive correlation between purchases of 'Semi-skimmed milk' and 'Liquid wholemilk, full price'.
(i)	Identifies regions C and E as outliers	AO3.2b	E1	Regions C and E do not follow the same pattern as the other regions in terms of purchases of skimmed milk and wholemilk or regions C and E can be excluded as they appear to be outliers.
ALT	Allow 1 mark if candidate states: weak negative correlation between consumption of 'Semi-skimmed milk' and 'Liquid wholemilk, full price' .	AO2.5	B1	
		AO3.2b	E0	
(ii)	States that there is evidence of negative correlation between purchases of 'Mineral or spring water' and 'Skimmed milk'.	AO3.1b	B1	Scatter graph indicates evidence of negative correlation between consumption of 'Mineral or spring water' and 'Skimmed milks'.
	The correlation is between water and milk purchases within the regions and not between purchases made by individual people so Bilal's claim is not proven	AO2.3	E1	Bilal's claim is not supported because the correlation in figure 2 is between water and milk purchases within the regions and he is assuming that individuals will follow a similar pattern but the data does not tell us anything about individuals, it refers to average purchases made within the nine regions.
(c)	Identifies London	AO2.2b	B1	London

Gives valid reason based on knowledge of the LDS	AO2.4	E1	Having studied the LDS I have identified a clear trend: London is a frequent outlier in most categories
Total 7 marks			

Q5.

- (a) (i) Mode = **253**
CAO

B1

1

- (ii) Median = **252**
CAO

B1

Upper quartile = **253**

Lower quartile = **250**

CAO; either

May be implied by $IQR = 3$

B1

Interquartile range = **3**

CAO; do not award if **seen** to be not based on 253 and 250

B1

3

- (b) Mean, $\bar{x} = \mathbf{251 \text{ to } 251.4}$
Award B1 if divisor **seen** not to be 85 but answer in range

$$AWFW \quad \sum fx = 21352 \quad \bar{x} = 251.2$$

B2

Note:

If B0 then can award

M1 for attempt at $\sum fx \div 85$ **seen**

*Ignore notation and condone incorrect midpoints
(eg upper or lower limits used)*

Standard deviation, s or $\sigma = \mathbf{4.21 \text{ to } 4.28}$

Award B1 if divisor **seen** not to be 84 or 85 but answer in range

$$AWFW \quad \sum fx^2 = 5365134$$

$$\sigma = 4.217$$

$$s = 4.242$$

B2

4

- (c) Interquartile range (IQR)
Named

B1

Not affected by unknown/large/small/extreme/outlying/227 & 271 values
Or equivalent
Dependent on previous B1
Only negative comments on other measures $\Rightarrow$ Bdep0

Bdep1

OR

More than one named $\Rightarrow$ B0 Bdep0
Range $\Rightarrow$ B0 Bdep0

Standard deviation (s or σ)
Named

(B1)

Uses all data values
Or equivalent
Dependent on previous (B1)
Only negative comments on other measures $\Rightarrow$ Bdep0
(Bdep1)

2

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Q6.

- (a) (i) Mode = **253**
CAO

B1

1

- (ii) Median = **252**
CAO

B1

Upper quartile = **253**

Lower quartile = **250**

CAO; either
May be implied by IQR = 3

B1

Interquartile range = 3

CAO; do not award if **seen** to be not based on 253 and 250

B1

3

- (b) (i) Range = 271 – 227 = 44

CAO; do not award if **seen** to be not based on 271 and 227

B1

1

- (ii) Mean, $\bar{x} = 251 \text{ to } 251.4$

Award B1 if divisor **seen** not to be 85 but answer in range

$$\text{AWFW } \sum fx = 21352 \quad \bar{x} = 251.2$$

B2

Note:

If B0 then can award

M1 for attempt at $\sum fx \div 85$ **seen**

Ignore notation and condone incorrect midpoints
(eg upper or lower limits used)

Standard deviation, s or $\sigma = 4.21 \text{ to } 4.28$

Award B1 if divisor **seen** not to be 84 or 85 but answer in range

$$\text{AWFW } \sum fx^2 = 5365134$$

$$\sigma = 4.217 \quad s = 4.242$$

B2

4

- (c) Interquartile range (IQR)

Named

B1

Not affected by unknown/large/small/extreme/outlying/227 & 271 values

Or equivalent

Dependent on previous B1

Only negative comments on other measures $\Rightarrow$ Bdep0

Bdep1

OR

More than one named $\Rightarrow$ B0 Bdep0

Range $\Rightarrow$ B0 Bdep0

Standard deviation (s or σ)
Named

(B1)

Uses all data values

Or equivalent

Dependent on previous (B1)

Only negative comments on other measures $\Rightarrow$ Bdep0

(Bdep1)

2

[11]

Q7.

- (a) y increases rapidly with x for small values of x and then levels out.

y increase with x

E1

Relationship changes

E1

Description of change

E1

3

- (b) Not appropriate because relationship nonlinear

Non-linear

M1

Not appropriate

A1

2

[5]